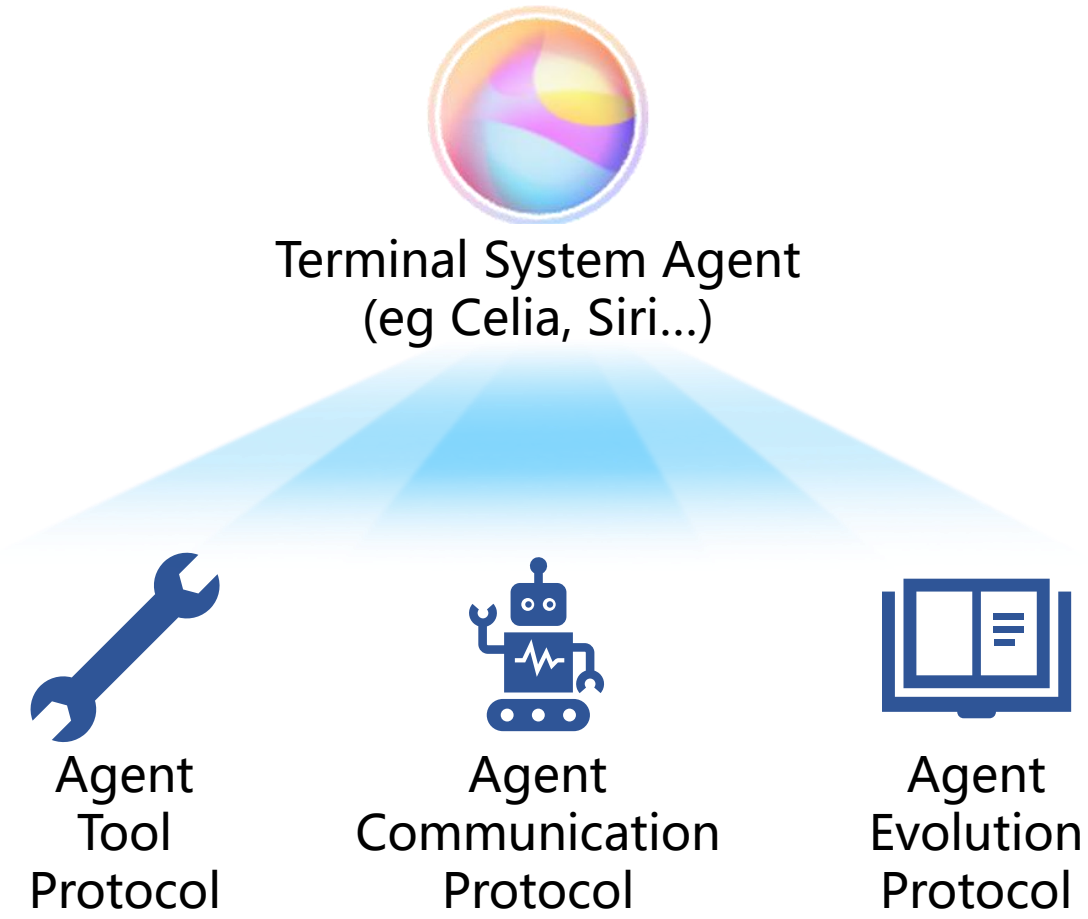
Agent-to-Agent/APPs/Tools Interaction and Invocation Scenarios on Terminals

Jinyang Li, Shuping Peng
lijinyang9@huawei.com; pengshuping@huawei.com

IETF 125 - Use Cases for Agentic AI Protocols Side Meeting

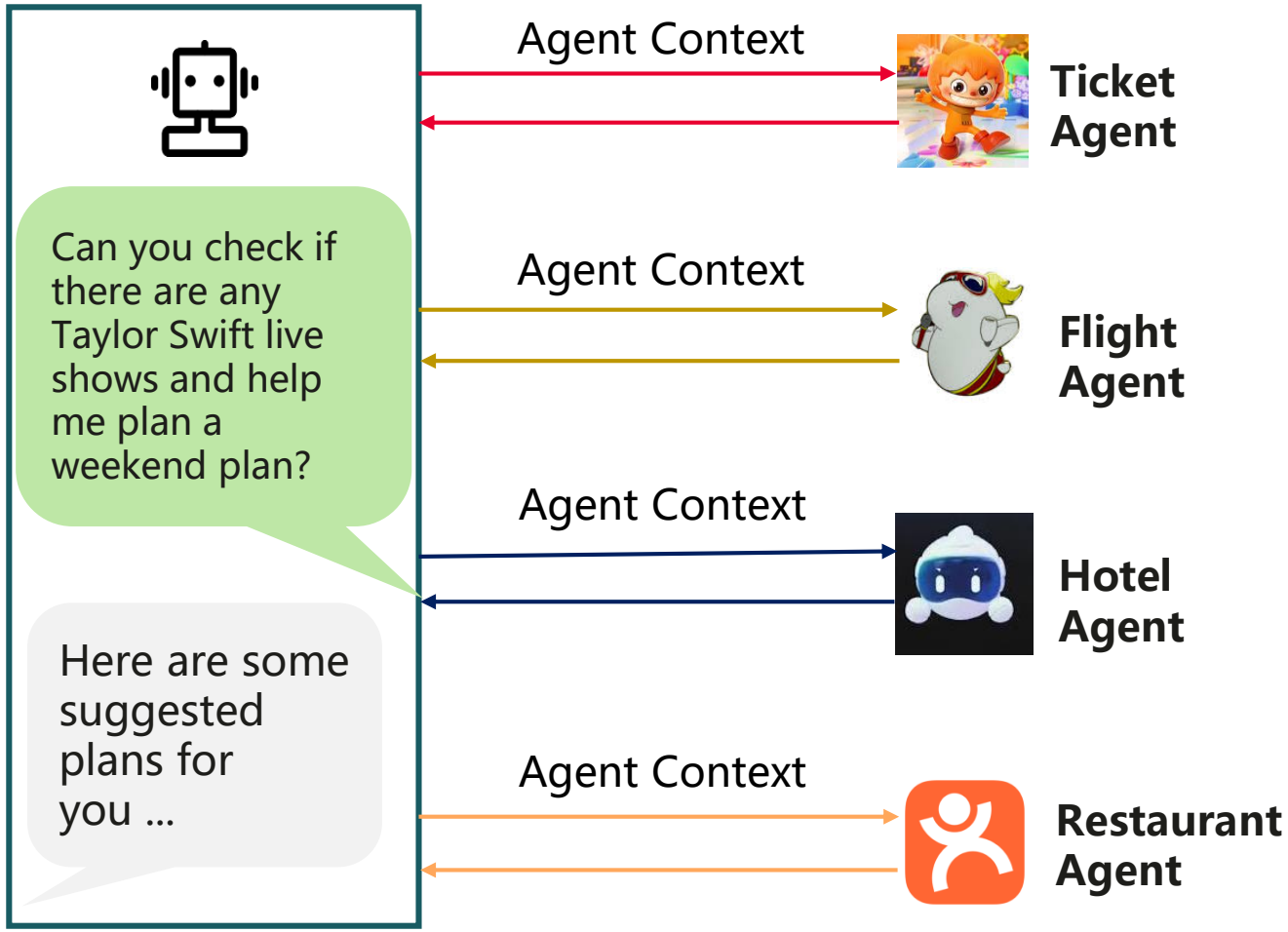
Overview: Agentic Protocols Use Cases on Terminal



- **Scenarios:**
  1. **Agent Communication Protocols**: To collaborate seamlessly with App Agents (e.g., A2A).
  2. **Agent Tool Protocols**: To utilize and execute agent tools (e.g., MCP).
  3. **Agent Evolution Protocols**: To continuously learn and adapt to user preferences over time.
- **Challenges:**
  1. Context synchronization across agents in multi-agent collaboration scenarios.
  2. Complex authz management within multi-step tool invocation workflows.
  3. Learning and evolving agent usage patterns from agent/tool interaction histories.

Use Case #1: Cross-App Agent Context Synchronization

(Agent Communication Protocol)



- **Scenario:** A system agent delegates a complex user request with multiple App Agents (e.g., travel planning).
- **Problem**: context is exchanged in plain text without structured semantics or cross-agent task state management.
- **Challenges**:
  - Excessive token consumption in high-concurrency scenarios.
  - Longer task execution time and reduced overall efficiency.
  - Lower task completion success rates.
- **Proposed draft**: [draft-chang-agent-context-interaction-01 - Agent Context Interaction Optimizations]
  - Context sync via structured context schema
  - Task management mechanism

Use Case #2: AuthZ in Multi-step AI agent Workflow

(Agent Tool Protocol)

- **Scenario**: A system AI agent plans a workflow that spans across multiple OAuth protected resources with distinct RS and scopes
- **Problems**: Agent protocols (e.g. MCP) adopt OAuth 2.0 AuthZ flow in a reactive and challenge-triggered way
- **Challenges**:
  - User is required to respond to multiple consent prompts, causing user friction and fatigue
  - Missing a single AuthZ requirement -> mid-workflow failure
- **Proposed draft**: [draft-jia-oauth-scope-aggregation-00 - OAuth 2.0 Scope Aggregation for Multi-Step AI Agent Workflows]
  - Records security requirement in resource metadata
  - Aggregate AuthZ scope and initiate a single AuthZ flow

Use Case #3: Experience Template Generation

(Agent Evolution Protocol)

- **Scenario:** Executing recurring complex tasks that require multi-turn dialogues to elicit user preferences and task parameters
- **Problem:** Existing protocols lack evolution mechanisms to extract reusable "experience templates" from history for future one-shot information gathering.
- **Challenges:**
  - Severe user fatigue from repetitive questioning.
  - Excessive token consumption and latency from redundant conversational turns.
  - Static agents failing to adapt continuously to user habits.
- **Draft WIP**
  - Abstract task-oriented experience template from interaction history

Conclusion

- **Summary**: Terminal Agent faces strict constraints in tokens, latency, and user friction. Existing stateless, plain-text, static protocols are insufficient to support complex, multi-agent workflows effectively.

- **Future work**: Shift towards structured, stateful, and self-evolving AI protocols.
  - Structured Context Sync: draft-chang-agent-context-interaction-01
  - Aggregated AuthZ: draft-jia-oauth-scope-aggregation-00
  - Agent Evolution: Experience Templates (Draft WIP)